Effective Nuclear Charge and Shielding

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

How hard does the nucleus really pull on an outer electron?

Targets

- I can estimate Z_eff as Z minus core electrons and use it to compare how hard atoms hold electrons.

Concepts

- **Core electrons** (core = Z – outer electrons): Every electron that is not in the outer ring.
- **Shielding** (a core electron cancels one proton): Core electrons cancel part of the nucleus's pull.
- **Z_eff** (Z_eff = Z – core): The pull an outer electron really feels. Estimate: Z minus the core electrons.
- **Distance** (pull = Z_eff ÷ n^2): The farther the ring, the weaker the pull (slot 7).

What the class predicted, and what is true

- Lithium and sodium each have one outer electron. Which atom holds its outer electron harder? **Lithium.** Sodium's outer electron is in ring 3, lithium's in ring 2. The farther electron is pulled less.

Huddle notes (answers filled in)

Definitions

- **Core electrons:** The electrons **inside** the outer ring.
- **Shielding:** Core electrons **cancel** part of the nucleus's pull.
- **Z_eff:** Z_eff = Z minus the **core** electrons.

Worked example

Sodium (Z = 11): how hard is its outer electron held?

- Rings: 2, 8, 1. Total **11**.
- Core = 2 + 8 = **10**.
- Z_eff = 11 – 10 = **1**.
- Pull = 1 ÷ 3^2 = **1/9**.

Label the sodium atom

1. The nucleus has charge +**11**.
2. The **core** rings hold **10** electrons.
3. The **outer** ring holds **1** electron.

One worked sample of each skill

Count the core electrons

Level 1. How many core electrons does this sodium atom have? Core electrons are the ones not in the outer ring.
Answer: 10 Rings: 2, 8, 1. The outer ring is not core, so 2 + 8 = 10.

Level 2. A neutral magnesium atom has Z = 12 and its electrons sit in rings of 2, 8, 2. How many are core electrons?
Answer: 10 The last ring (2) is the outer ring. Core: 2 + 8 = 10.

Level 3. A neutral atom has Z = 19 and 1 electron in its outer ring. How many core electrons does it have?
Answer: 18 19 – 1 = 18 core electrons.

Level 4. A neutral atom has Z = 18 and 8 electrons in its outer ring. How many core electrons does it have?
Answer: 10 18 – 8 = 10 core electrons.

Estimate Z_eff

Level 1. The picture shows sodium (Z = 11). Z_eff is Z minus the core electrons. What is Z_eff for sodium?
Answer: 1 11 – 10 = 1.

Level 2. Estimate Z_eff for sodium (Z = 11, rings of 2, 8, 1).
Answer: 1 Core = 10. Z_eff = 11 – 10 = 1.

Level 3. An atom in period 3 has Z_eff = 1. Which atom is it?
(A) Phosphorus **(B) Sodium** (C) Chlorine (D) Aluminum
Answer: B Z_eff = Z – 10, so Z = 1 + 10 = 11: sodium.

Level 4. An atom in period 3 has Z_eff = 8. Which atom is it?
(A) Argon (B) Silicon (C) Magnesium (D) Chlorine
Answer: A Z_eff = Z – 10, so Z = 8 + 10 = 18: argon.

Which holds its outer electron harder

Level 1. Each atom has one outer electron. Which atom holds its outer electron harder, lithium (left) or potassium (right)?
(A) Lithium (B) Potassium (C) They hold it equally
Answer: A Both have Z_eff = 1. Lithium's outer electron is in ring 2 and potassium's is in ring 4, so lithium pulls harder.

Level 2. Which atom holds its outer electron harder, calcium (Z = 20) or magnesium (Z = 12)? Find Z_eff for each.

(A) Calcium **(B) Magnesium** (C) They hold it equally
Answer: B Both have Z_eff = 2. Magnesium's outer electron is in ring 3 and calcium's is in ring 4, so magnesium pulls harder.

Level 3. Which atom holds its outer electron harder, silicon (Z = 14) or calcium (Z = 20)? Use Z_eff and the outer ring number.
(A) Calcium **(B) Silicon** (C) They hold it equally
Answer: B Silicon: $4 \div 3^2 = 4/9$. Calcium: $2 \div 4^2 = 2/16$. The larger number pulls harder.

Level 4. Which atom holds its outer electron harder, argon (Z = 18) or silicon (Z = 14)? Use Z_eff and the outer ring number.
(A) They hold it equally (B) Silicon **(C) Argon**
Answer: C Both outer electrons are in ring 3. Argon has Z_eff = 8 and silicon has 4, so argon pulls harder.